Design & Development of Wrist Mounted Smart Device with Integrated GUI .



Abstract

A user-friendly device with a larger touch screen than the average smart watch, making it easier for consumers to interact with and a multitude of functionality, updatability, and possibility. Creating an enhanced user experience that would feel like having all the capabilities of a desktop computer or smart phone with the ease and convenience of a hands-free device, with the increased size of the device in comparison to your average smart watch, it would allow for more advanced & versatile pieces of hardware, i.e. increased storage and processing power as the main features. The design and development of the device will incorporate many different elements, from the physical look of the product to the layouts of the graphical user interfaces with the possibility of its proposed multi-functionality, being considered for a variety of uses in the business world, medical sector, and personal convenience.

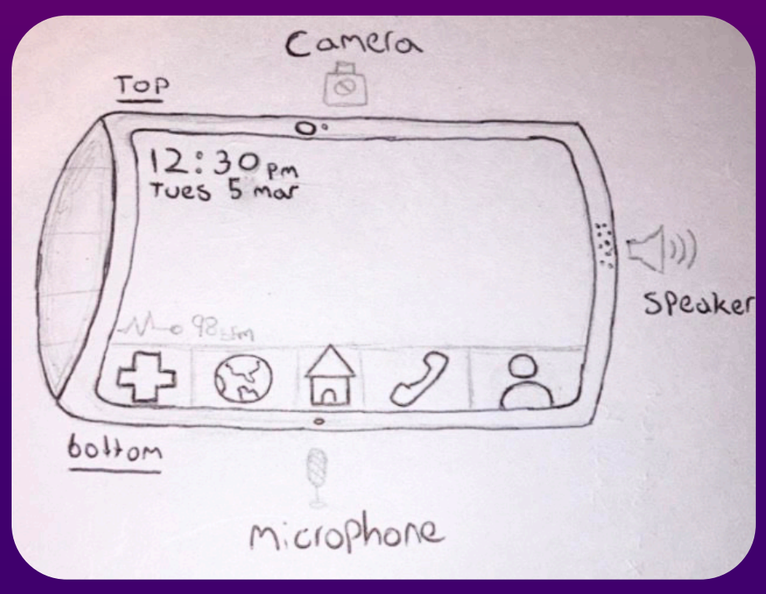
Project Aims

- **Improve user experience when interacting with wrist mounted smart device.**
- **Research & identify possible opportunities and implementation of device i.e. personal convenience, assistive technology, military/medical application...**
- **Design aesthetically pleasing graphical user-interface for the devices 3.5-inch screen.**
- **Develop wrist mounted smart device prototype with initial features and potential future modifications.**

Product

Pictured on the right is a sketched outline of what the envisioned product will hopefully look like:

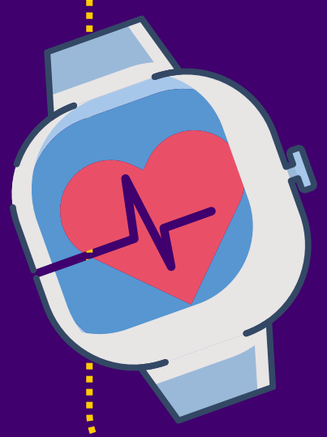
- **3.5 inch touchscreen**
- **Microphone**
- **Camera & Light**
- **Sensors**



Seen along the bottom of the screen is the main user menu, consisting of several options:

- **Home Screen**
- **Health**
- **Internet**
- **Phone**
- **Profile/Settings**

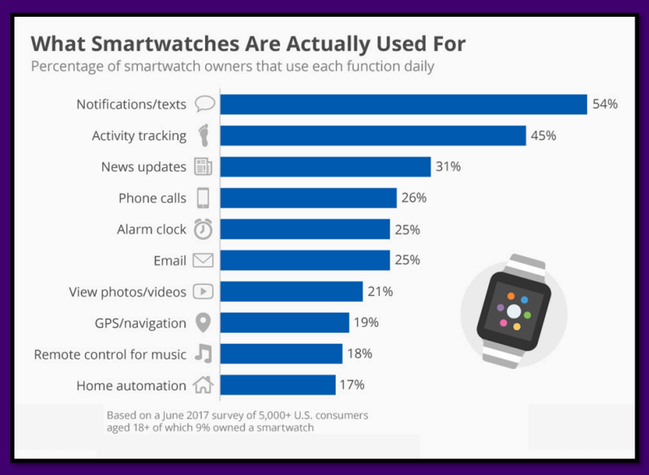
Sensors



- Pressure
- Temperature
- Compass
- Oximetry
- GPS
- Gyroscope
- Heart Rate Monitor
- Accelerometer

there are a variety of potential sensors to be considered for implementation of the device, each one accompanied by its own intended purpose and the difficulties associated with incorporating it into the final product.

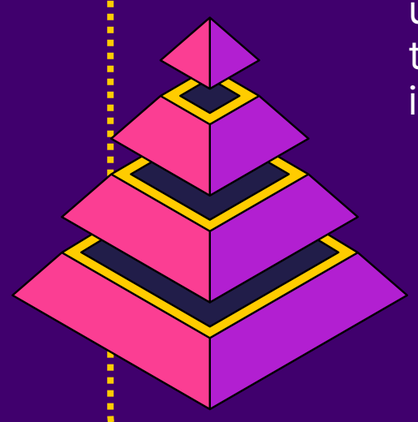
Market Comparison



Market Share:

Apple (36%),
Samsung (11%),
Huawei (7%),
Imoo (3%),
Garmin (3%)

Methodology

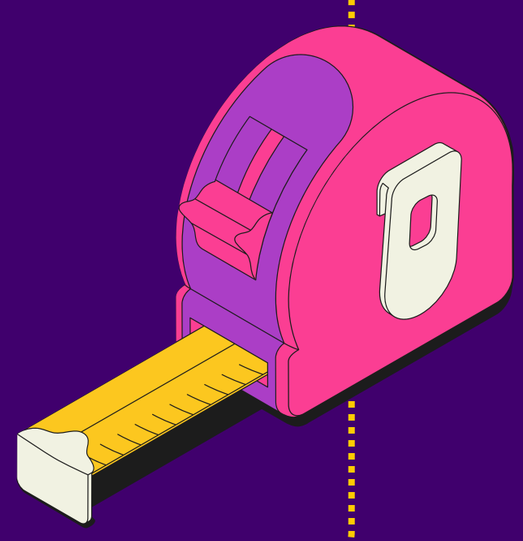


utilizing the waterfall methodology in a top down fashion to develop the project in a step by step manner.

- Requirements & Research
- Design
- Implementation
- Testing
- Deployment
- Maintenance

Design

1. defining the aesthetic of the device and its application layouts, featuring an original operating system built from the ground up with inspirations from current market trends and what users and customers find preferential.
2. built on top of a modified ubuntu system using a raspberry Pi.



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